

Paper Summary

1. Problem Understanding

- Definition of Autism Spectrum Disorder (ASD)
- Importance of early diagnosis
- Screening vs clinical diagnosis

2. Dataset Details

- Dataset Name and Source
- Number of samples
- Age group (toddlers / children / adults)
- Class balance (ASD vs Non-ASD ratio)
- Type of data used (Questionnaire data/Behavioral data/Eye-tracking/Speech data/MRI/fMRI imaging)

3. Features Used

- Behavioral features (AQ-10, AQ-Child)/Social interaction indicators/ Speech/voice features/ Brain imaging biomarkers/ Demographic factors (age, gender)
- Feature selection methods.
- Dimensionality reduction (PCA, etc.)

4. Machine Learning Models Compared

- Logistic Regression, SVM, Random Forest, KNN, XGBoost, Deep Learning (CNN, LSTM) etc.

Study:

- Which model performs best?
- Why does it perform better?
- Model complexity vs dataset size

5. Evaluation Metrics Used

- Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC-ROC etc..

6. Experimental Setup

- Train-test split ratio
- Cross-validation method
- Hyperparameter tuning
- Handling class imbalance (SMOTE, etc.)

7. Key Contributions of the Paper

- Novel feature engineering?
- Hybrid or ensemble models?
- Reduced screening time?
- Improved clinical usability?

8. Limitations of the paper

- Small dataset size, No external validation, Class imbalance, Lack of explainability, High computational cost, etc..

Authors	Year	Problem/Title	Dataset	Model	Performance matrices	Contribution	Limitation